

EXPERIMENT-5

Simulation of logic circuits using Transistor Level Modelling

Objective:

To understand the concepts related to structural modeling style and write Verilog programs using the same.

Theory:

In structural modelling, the digital system is described using the interconnected components or modules, essentially representing the hardware structure of the system. It closely resembles the actual circuit where logic gates and other hardware modules are interconnected to form the final system. Structural modelling is a lower level of abstraction compared to behavioural modelling, and it describes how components are connected rather than how they behave.

In Verilog, the structural model is defined using module instances and port connections. Modules can be instantiated and connected through nets to represent wires.

NMOS and PMOS Transistors:

- **NMOS Transistor:** Conducts when the gate input is high (1), creating a connection between the source and drain.
- **PMOS Transistor:** Conducts when the gate input is low (0), connecting the source and drain.

Syntax for NMOS and PMOS Transistors:

In Verilog, switch-level modelling is defined using nmos and pmos keywords. The syntax for instantiating these transistors is:

nmos (drain, source, gate); // For NMOS transistor

pmos (drain, source, gate); // For PMOS transistor

- **Drain:** The output of the transistor.
- **Source:** The source of the transistor (ground for NMOS or vdd for PMOS).
- **Gate:** The control signal for the transistor (determines whether the transistor is on / off).

Exercise Problems

1. Write the Verilog code and testbenches for 3 input CMOS NAND gate

Circuit diagram:

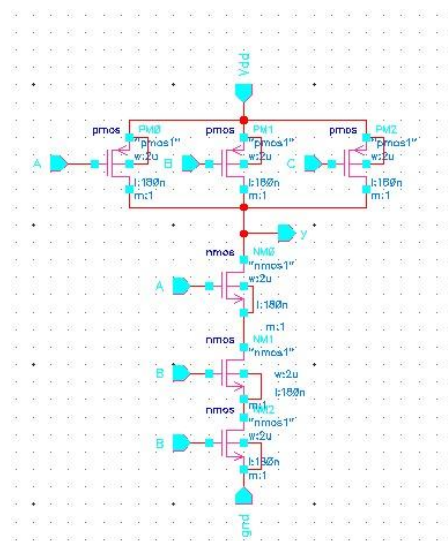


Figure 1: CMOS circuit diagram of three input NAND gate.

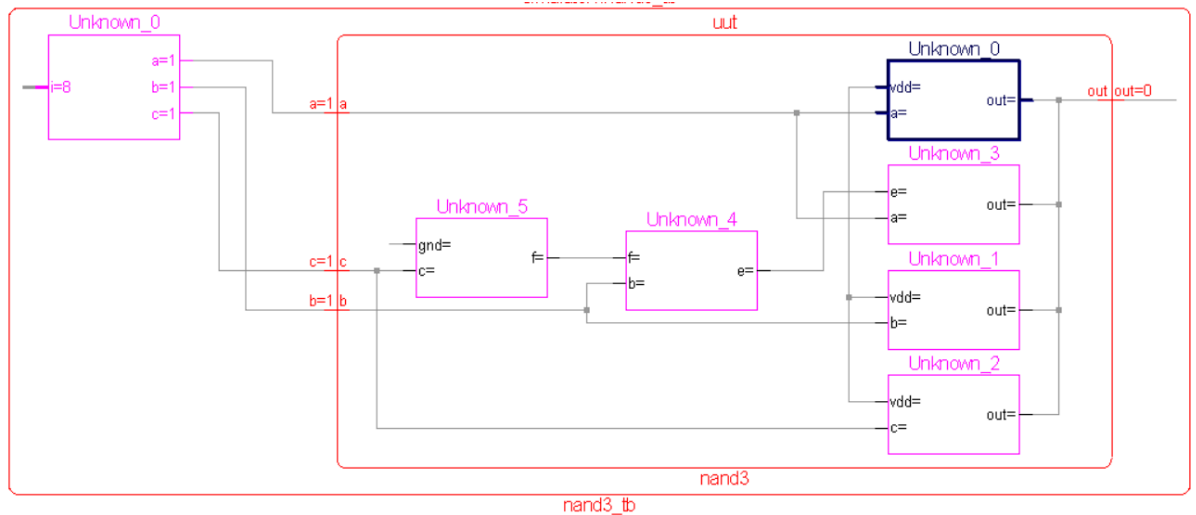


Figure 2: Circuit diagram of three input NAND gate.

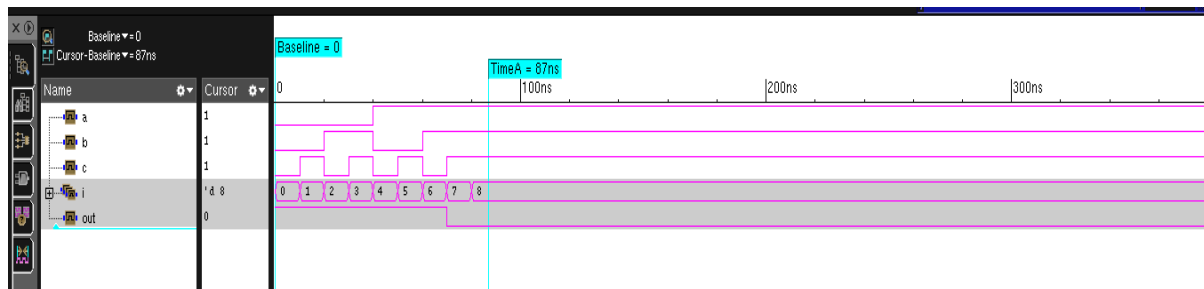
Source code:

```
module nand3(out,a,b,c);
output out;
input a,b,c;
supply1 vdd;
supply0 gnd;
pmos m1(out,vdd,a);
pmos m2(out,vdd,b);
pmos m3(out,vdd,c);
nmos m4(out,e,a);
nmos m5(e,f,b);
nmos m6(f,gnd,c);
endmodule
```

Testbench code:

```
module nand3_tb();
reg a,b,c;
wire out;
integer i;

nand3 uut (out,a,b,c);
initial begin
$monitor("Time = %0t, a = %0b, b = %0b, c = %0b, out = %0b ",$time,a,b,c,b);
a=0;b=0;c=0;
for(i=0;i<=7;i=i+1)
begin
{a,b,c}=i;
#10;
end
#300 $stop;
end
endmodule
```

Simulation and Console results:**Figure 3: Simulation result of 3 input NAND gate.**

```

ncsim>
ncsim> source /home/install/INCISIVE152/tools/inca/files/ncsimrc
ncsim> database -open waves -into waves.shm -default
Created default SHM database waves
ncsim> probe -create -shm nand3_tb.a nand3_tb.b nand3_tb.c nand3_tb.i nand3_tb.out
Created probe 1
ncsim> run
Time = 0, a = 0, b = 0, c = 0, out = 0
Time = 10, a = 0, b = 0, c = 1, out = 0
Time = 20, a = 0, b = 1, c = 0, out = 1
Time = 30, a = 0, b = 1, c = 1, out = 1
Time = 40, a = 1, b = 0, c = 0, out = 0
Time = 50, a = 1, b = 0, c = 1, out = 0
Time = 60, a = 1, b = 1, c = 0, out = 1
Time = 70, a = 1, b = 1, c = 1, out = 1
Simulation stopped via $stop(1) at time 380 NS + 0
ncsim>

```

Figure 4: Console result of 3 input nand gate.**2. Write the Verilog code and testbenches for $F = \sim E(A + BCD)$:****Truth table:**

A	B	C	D	E	$F = \sim E(A + BCD)$
0	0	0	0	0	0
0	1	1	1	0	1
1	0	0	0	0	1
1	1	1	1	0	1
0	0	1	1	1	0
0	1	1	0	0	0
1	1	0	1	1	0
1	0	1	0	0	1
0	1	0	0	1	0
1	1	1	1	1	0

Table 1: Truth table for the following expression.

Circuit Diagram:

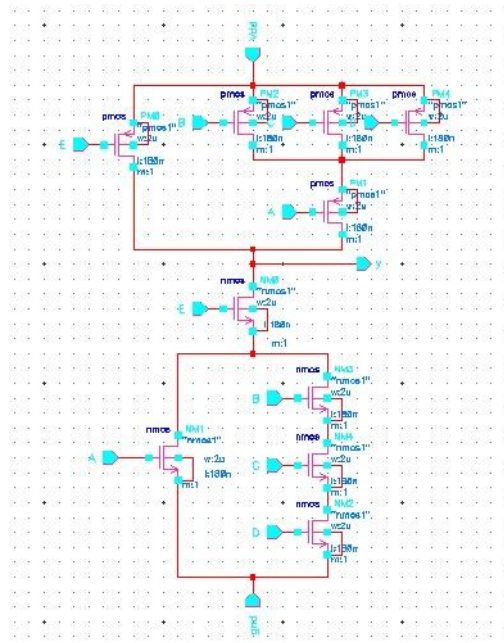


Figure 5: CMOS circuit diagram.

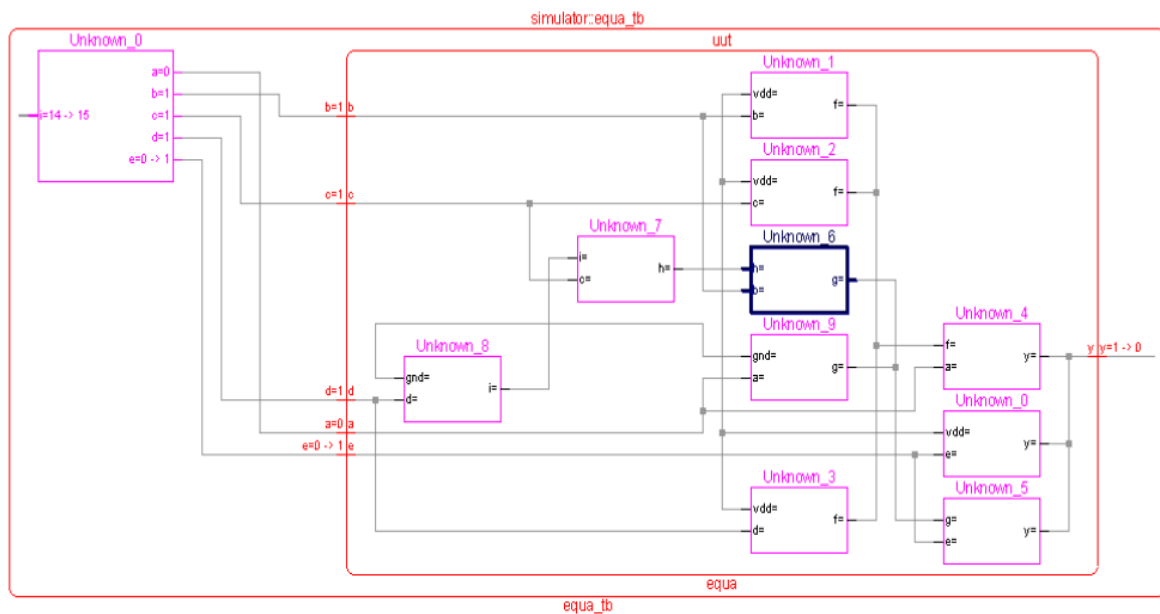


Figure 6: Circuit diagram.

Source code:

```
module equa(a,b,c,d,e,y);
input a,b,c,d,e;
output y;
supply1 vdd;
supply0 gnd;

pmos m1(y,vdd,e);
pmos m2(f,vdd,b);
pmos m3(f,vdd,c);
```

```

pmos m4(f,vdd,d);
pmos m5(y,f,a);

nmos m6(y,g,e);
nmos m7(g,h,b);
nmos m8(h,i,c);
nmos m9(i,gnd,d);
nmos m10(g,gnd,a);
endmodule

```

Testbench code:

```

module equa_tb();
reg a,b,c,d,e;
wire y;
integer i;
equa uut (a,b,c,d,e,y);
initial begin
$monitor("Time = %0t, a = %0b, b = %0b, c = %0b, d = %0b, e = %0b y = %0b",
$time,a,b,c,d,e,y);
a=0;b=0;c=0;d=0;e=0;
for(i=0;i<=32;i=i+1)
begin
{a,b,c,d,e}=i;
#10;
end
#300 $stop;
end
endmodule

```

Simulation and console output:



Figure 7: Simulation waveform of the above expression.

```

Created probe 1
ncsim> run
Time = 0, a = 0, b = 0, c = 0, d = 0, e = 0 y = 1
Time = 10, a = 0, b = 0, c = 0, d = 0, e = 1 y = 1
Time = 20, a = 0, b = 0, c = 0, d = 1, e = 0 y = 1
Time = 30, a = 0, b = 0, c = 0, d = 1, e = 1 y = 1
Time = 40, a = 0, b = 0, c = 1, d = 0, e = 0 y = 1
Time = 50, a = 0, b = 0, c = 1, d = 0, e = 1 y = 1
Time = 60, a = 0, b = 0, c = 1, d = 1, e = 0 y = 1
Time = 70, a = 0, b = 0, c = 1, d = 1, e = 1 y = 1
Time = 80, a = 0, b = 1, c = 0, d = 0, e = 0 y = 1
Time = 90, a = 0, b = 1, c = 0, d = 0, e = 1 y = 1
Time = 100, a = 0, b = 1, c = 0, d = 1, e = 0 y = 1
Time = 110, a = 0, b = 1, c = 0, d = 1, e = 1 y = 1
Time = 120, a = 0, b = 1, c = 1, d = 0, e = 0 y = 1
Time = 130, a = 0, b = 1, c = 1, d = 0, e = 1 y = 1
Time = 140, a = 0, b = 1, c = 1, d = 1, e = 0 y = 1
Time = 150, a = 0, b = 1, c = 1, d = 1, e = 1 y = 0
Time = 160, a = 1, b = 0, c = 0, d = 0, e = 0 y = 1
Time = 170, a = 1, b = 0, c = 0, d = 0, e = 1 y = 0
Time = 180, a = 1, b = 0, c = 0, d = 1, e = 0 y = 1
Time = 190, a = 1, b = 0, c = 0, d = 1, e = 1 y = 0
Time = 200, a = 1, b = 0, c = 1, d = 0, e = 0 y = 1
Time = 210, a = 1, b = 0, c = 1, d = 0, e = 1 y = 0
Time = 220, a = 1, b = 0, c = 1, d = 1, e = 0 y = 1
Time = 230, a = 1, b = 0, c = 1, d = 1, e = 1 y = 0
Time = 240, a = 1, b = 1, c = 0, d = 0, e = 0 y = 1
Time = 250, a = 1, b = 1, c = 0, d = 0, e = 1 y = 0
Time = 260, a = 1, b = 1, c = 0, d = 1, e = 0 y = 1
Time = 270, a = 1, b = 1, c = 0, d = 1, e = 1 y = 0
Time = 280, a = 1, b = 1, c = 1, d = 0, e = 0 y = 1
Time = 290, a = 1, b = 1, c = 1, d = 0, e = 1 y = 0
Time = 300, a = 1, b = 1, c = 1, d = 1, e = 0 y = 1
Time = 310, a = 1, b = 1, c = 1, d = 1, e = 1 y = 0
Time = 320, a = 0, b = 0, c = 0, d = 0, e = 0 y = 1
Simulation stopped via $stop(1) at time 630 NS + 0
ncsim>

```

Figure 8: Console waveform of the above expression.

Exercise Problems

1. Write Verilog code for the following combinational logic function using CMOS logic
a) $F = \sim(ABC + DE)$, b) XOR gate, c) half adder
2. Write Verilog code for the full adder and verify the output.
3. Write Verilog code for the full subtractor and verify the output.
4. Write Verilog code for the 4:1 MUX and verify the output.
5. Write Verilog code for the 2:4 decoder and verify the output.